

EDEXCEL INTERNATIONAL A LEVEL

WMA11/01 January 2024 Worked Solutions

Jan 2024 paper rebuilt with the worked solution after each question.

10

paper questions

WMA11

Pure 1

**Single-
paper
solutions**standalone IAL
route**Dr Eslam Ahmed**

Prepared for Dr Eslam Ahmed - eliteigcse.com

P1

PAST PAPER

WMA11/01 January 2024

Jan 2024 | 10 questions | 75 marks

10

questions

75

marks

Answers

worked solution
after each
question

Question 1

Integration

1. Find

$$\int (2x - 5)(3x + 2)(2x + 5)dx$$

writing your answer in simplest form.

(5)

(Total for Question 1 is 5 marks)

Worked Solution - Question 1

1. Expand two brackets first

$$(2x - 5)(3x + 2) = 6x^2 - 11x - 10.$$

2. Multiply by the third bracket

$$(6x^2 - 11x - 10)(2x + 5) = 12x^3 + 8x^2 - 75x - 50.$$

3. Integrate term by term

$$\int (12x^3 + 8x^2 - 75x - 50) dx = 3x^4 + \frac{8}{3}x^3 - \frac{75}{2}x^2 - 50x + c.$$

Final answer

$$3x^4 + \frac{8}{3}x^3 - \frac{75}{2}x^2 - 50x + c.$$

Question 2

Basic Trigonometry

2. The triangle ABC is such that

- $AB = 15 \text{ cm}$
- $AC = 25 \text{ cm}$
- $\text{angle } BAC = \theta^\circ$
- $\text{area triangle } ABC = 100 \text{ cm}^2$

(a) Find the value of $\sin \theta^\circ$

(2)

Given that $\theta > 90$

(b) find the length of BC , in cm, to 3 significant figures.

(3)

(Total for Question 2 is 5 marks)

Worked Solution - Question 2

Topic group

1. Use the triangle area formula

$$100 = \frac{1}{2}(15)(25) \sin \theta.$$

2. Find sine theta

$$\sin \theta = \frac{200}{375} = \frac{8}{15}.$$

3. Use theta greater than 90 degrees

Since $\theta > 90^\circ$, $\cos \theta$ is negative. So $\cos \theta = -\sqrt{1 - \left(\frac{8}{15}\right)^2} = -\frac{\sqrt{161}}{15}.$

4. Apply the cosine rule

$$BC^2 = 15^2 + 25^2 - 2(15)(25) \cos \theta = 850 + 50\sqrt{161}.$$

5. Round the length

$$BC = \sqrt{850 + 50\sqrt{161}} \approx 38.5 \text{ cm to 3 significant figures.}$$

Final answer

(a) $\sin \theta = \frac{8}{15}.$

(b) $BC = 38.5 \text{ cm.}$

Question 3

Differentiation

3. The curve C has equation

$$y = \frac{5x^3 - 8}{2x^2} \quad x > 0$$

- (a) Find $\frac{dy}{dx}$ writing your answer in simplest form.

(4)

The point $P(2, 4)$ lies on C .

- (b) Find an equation for the tangent to C at P writing your answer in the form $ax + by + c = 0$, where a , b and c are integers.

(3)

(Total for Question 3 is 7 marks)

Worked Solution - Question 3

1. Rewrite y

$$y = \frac{5x^3 - 8}{2x^2} = \frac{5}{2}x - 4x^{-2}.$$

2. Differentiate

$$\frac{dy}{dx} = \frac{5}{2} + 8x^{-3} = \frac{5}{2} + \frac{8}{x^3}.$$

3. Find the gradient at P

$$\text{At } x = 2, \frac{dy}{dx} = \frac{5}{2} + \frac{8}{8} = \frac{7}{2}.$$

4. Use point-gradient form

$$\text{The tangent through } P(2, 4) \text{ is } y - 4 = \frac{7}{2}(x - 2).$$

5. Rearrange

$$2y - 8 = 7x - 14, \text{ so } 7x - 2y - 6 = 0.$$

Final answer

$$(a) \frac{dy}{dx} = \frac{5}{2} + \frac{8}{x^3}.$$

$$(b) 7x - 2y - 6 = 0.$$

Question 4

4.

In this question you must show all stages of your working.

Solutions relying on calculator technology are not acceptable.

(a) By substituting $p = 2^x$, show that the equation

$$2 \times 4^x - 2^{x+3} = 17 \times 2^{x-1} - 4$$

can be written in the form

$$4p^2 - 33p + 8 = 0 \quad (3)$$

(b) Hence solve

$$2 \times 4^x - 2^{x+3} = 17 \times 2^{x-1} - 4 \quad (3)$$

(Total for Question 4 is 6 marks)

Worked Solution - Question 4

Topic group

1. Substitute p equals 2 to the x

If $p = 2^x$, then $4^x = p^2$, $2^{x+3} = 8p$, and $2^{x-1} = \frac{p}{2}$.

2. Rewrite the equation

$$2p^2 - 8p = 17\left(\frac{p}{2}\right) - 4.$$

3. Clear the fraction

Multiplying by 2 gives $4p^2 - 16p = 17p - 8$, so $4p^2 - 33p + 8 = 0$.

4. Solve the quadratic

$$4p^2 - 33p + 8 = (4p - 1)(p - 8), \text{ so } p = \frac{1}{4} \text{ or } p = 8.$$

5. Return to x

$$2^x = \frac{1}{4} = 2^{-2} \text{ gives } x = -2, \text{ and } 2^x = 8 = 2^3 \text{ gives } x = 3.$$

Final answer

$$x = 3 \text{ or } x = -2.$$

Question 5

Straight Line

5.

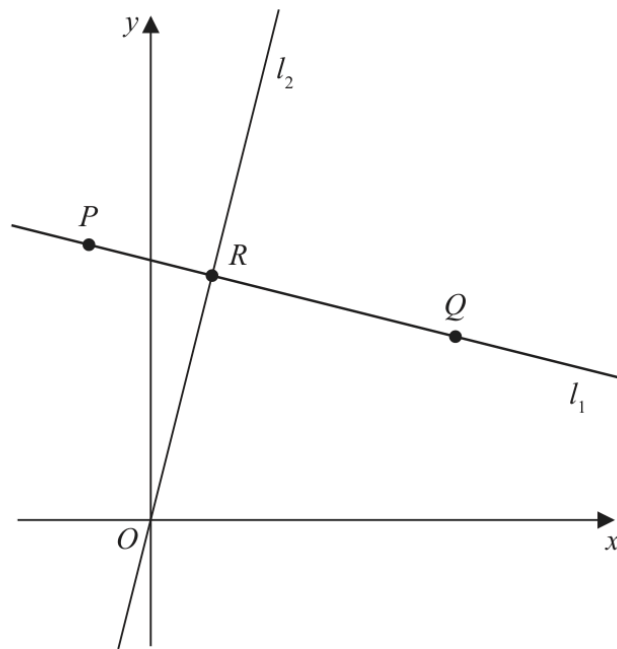


Figure 1

In this question you must show all stages of your working.

Solutions relying on calculator technology are not acceptable.

The straight line l_1 , shown in Figure 1, passes through the points $P(-2, 9)$ and $Q(10, 6)$.

- (a) Find the equation of l_1 , giving your answer in the form $y = mx + c$, where m and c are constants to be found.

(3)

The straight line l_2 passes through the origin O and is perpendicular to l_1 .

The lines l_1 and l_2 meet at the point R as shown in Figure 1.

- (b) Find the coordinates of R .

(4)

- (c) Find the exact area of triangle OPQ .

(3)

(Total for Question 5 is 10 marks)

Worked Solution - Question 5

1. Find the gradient of l_1

$$m = \frac{6 - 9}{10 - (-2)} = -\frac{3}{12} = -\frac{1}{4}.$$

2. Find the equation of l_1

Using $P(-2, 9)$, $y = -\frac{1}{4}x + c$. Then $9 = \frac{1}{2} + c$, so $c = \frac{17}{2}$.

3. Find the equation of l_2

A line perpendicular to gradient $-\frac{1}{4}$ has gradient 4. Since l_2 passes through the origin, l_2 is $y = 4x$.

4. Find R

At the intersection, $4x = -\frac{1}{4}x + \frac{17}{2}$. Hence $\frac{17}{4}x = \frac{17}{2}$, so $x = 2$ and $y = 8$.

5. Use coordinate area

For $O(0, 0)$, $P(-2, 9)$ and $Q(10, 6)$, the area is $\frac{1}{2}|(-2)(6) - 9(10)| = 51$.

Final answer

(a) $y = -\frac{1}{4}x + \frac{17}{2}$.

(b) $R = (2, 8)$.

(c) Area = 51.

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6 marks

Question 6

Trigonometric Functions

6.

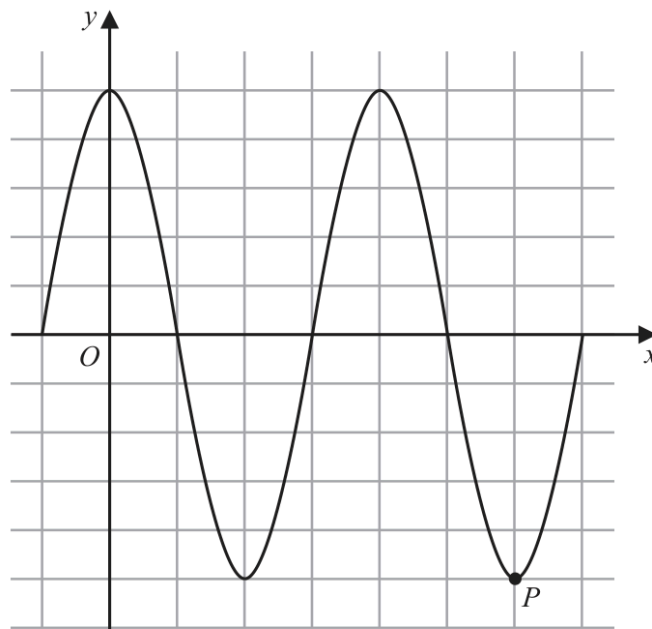


Figure 2

Figure 2 shows a plot of part of the curve C_1 with equation

$$y = 5 \cos x$$

with x being measured in degrees.

The point P , shown in Figure 2, is a minimum point on C_1

(a) State the coordinates of P

(2)

The point Q lies on a different curve C_2

Given that point Q

- is a maximum point on the curve
- is the maximum point with the **smallest** x coordinate, $x > 0$

(b) find the coordinates of Q when

(i) C_2 has equation $y = 5 \cos x - 2$

(ii) C_2 has equation $y = -5 \cos x$

(4)

(Total for Question 6 is 6 marks)

Worked Solution - Question 6

1. Use the cosine graph

For $y = 5 \cos x$, the minimum value is -5 and it occurs at $x = 180^\circ + 360^\circ n$.

2. Read P from the displayed minimum

The plotted point P is the displayed minimum at $x = 540^\circ$, so $P = (540^\circ, -5)$.

3. Translate the graph down

For $y = 5 \cos x - 2$, the maximum value is $5 - 2 = 3$. The smallest positive x -coordinate for a maximum shown is $x = 360^\circ$, so $Q = (360^\circ, 3)$.

4. Reflect in the x-axis

For $y = -5 \cos x$, a maximum occurs where $\cos x = -1$, first at $x = 180^\circ$. The maximum value is 5 , so $Q = (180^\circ, 5)$.

Final answer

(a) $P = (540^\circ, -5)$.

(b)(i) $Q = (360^\circ, 3)$.

(b)(ii) $Q = (180^\circ, 5)$.

Question 7

Graphs of Functions

7. (a) Sketch the graph of the curve C with equation

$$y = \frac{4}{x - k}$$

where k is a positive constant.

Show on your sketch

- the coordinates of any points where C cuts the coordinate axes
- the equation of the vertical asymptote to C

(4)

Given that the straight line with equation $y = 9 - x$ does not cross or touch C

- (b) find the range of values of k .

(5)

(Total for Question 7 is 9 marks)

Worked Solution - Question 7

1. Identify the asymptote

For $y = \frac{4}{x - k}$, the vertical asymptote is where the denominator is zero, so $x = k$.

2. Find axis intercepts

At $x = 0$, $y = -\frac{4}{k}$, so the y-intercept is $(0, -\frac{4}{k})$. The graph has no x-intercept because the numerator is never zero.

3. Set up intersection with the line

If the line $y = 9 - x$ meets the curve, then $9 - x = \frac{4}{x - k}$.

4. Form a quadratic

$(9 - x)(x - k) = 4$, which simplifies to $x^2 - (9 + k)x + 9k + 4 = 0$.

5. Use no real intersection

For the line not to cross or touch the curve, the discriminant must be negative.

6. Solve the inequality

$(9 + k)^2 - 4(9k + 4) < 0$, so $k^2 - 18k + 65 < 0$. Since $k^2 - 18k + 65 = (k - 5)(k - 13)$, the range is \$5

Final answer

(a) y-intercept $(0, -\frac{4}{k})$, no x-intercept, vertical asymptote $x = k$.

\$(b) \setminus 5

Question 8

Radians

8.

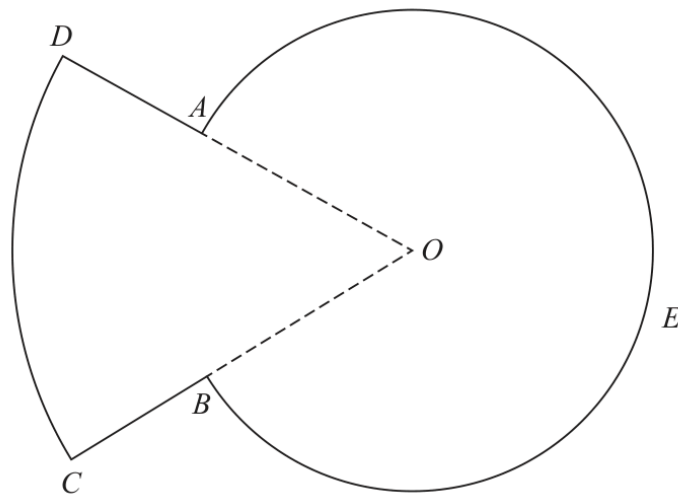


Figure 3

Figure 3 shows a sketch of the plan view of a platform.

The plan view of the platform consists of a sector DOC of a circle centre O joined to a sector $AOBEA$ of a different circle, also with centre O .

Given that

- angle $AOB = 0.8$ radians
- arc length $CD = 9$ m
- $DA:AO = 3:5$

(a) show that $AO = 7.03$ m to 3 significant figures.

(3)

(b) Find the perimeter of the platform, in m, to 3 significant figures.

(3)

(c) Find the total area of the platform, giving your answer in m^2 to the nearest whole number.

(3)

(Total for Question 8 is 9 marks)

Worked Solution - Question 8

1. Find the outer radius

For sector DOC , arc length $CD = 9$ and angle 0.8 radians, so

$$OD = \frac{9}{0.8} = 11.25 \text{ m.}$$

2. Use the ratio DA to AO

$DA : AO = 3 : 5$, so $OD = DA + AO$ represents 8 parts. Hence

$$AO = \frac{5}{8}(11.25) = 7.03125 \text{ m} \approx 7.03 \text{ m.}$$

3. Find the straight side lengths

$$DA = CB = OD - AO = 11.25 - 7.03125 = 4.21875 \text{ m.}$$

4. Find the major arc

The major sector $AOBEA$ has angle $2\pi - 0.8$, so arc

$$AEB = 7.03125(2\pi - 0.8).$$

5. Calculate the perimeter

$$\text{Perimeter} = 9 + 2(4.21875) + 7.03125(2\pi - 0.8) \approx 56.0 \text{ m.}$$

6. Calculate the total area

$$\text{Area} = \frac{1}{2}(11.25)^2(0.8) + \frac{1}{2}(7.03125)^2(2\pi - 0.8) \approx 186 \text{ m}^2 \text{ to the nearest whole number.}$$

Final answer

(a) $AO = 7.03 \text{ m}$.

(b) 56.0 m .

(c) 186 m^2 .

Question 9

Quadratics

9. The curve C_1 has equation $y = f(x)$.

Given that

- $f(x)$ is a quadratic expression
- C_1 has a maximum turning point at $(2, 20)$
- C_1 passes through the origin

- (a) sketch a graph of C_1 showing the coordinates of any points where C_1 cuts the coordinate axes,

(2)

- (b) find an expression for $f(x)$.

(3)

The curve C_2 has equation $y = x(x^2 - 4)$

Curve C_1 and C_2 meet at the origin, and at the points P and Q

Given that the x coordinate of the point P is negative,

- (c) using algebra and showing all stages of your working, find the coordinates of P

(5)

(Total for Question 9 is 10 marks)

Worked Solution - Question 9

1. Use symmetry for the sketch

The maximum is at $(2, 20)$ and the curve passes through $(0, 0)$. Since the axis of symmetry is $x = 2$, the other x-intercept is $(4, 0)$.

2. Use completed-square form

Let $f(x) = a(x - 2)^2 + 20$.

3. Use the origin

$0 = a(0 - 2)^2 + 20$, so $4a = -20$ and $a = -5$.

4. Write $f(x)$

$f(x) = -5(x - 2)^2 + 20 = -5x^2 + 20x = -5x(x - 4)$.

5. Equate the two curves

$-5x^2 + 20x = x(x^2 - 4) = x^3 - 4x$.

6. Factorise

$x^3 + 5x^2 - 24x = 0$, so $x(x^2 + 5x - 24) = 0$ and $x(x + 8)(x - 3) = 0$.

7. Use the negative x-coordinate

The negative x-coordinate is $x = -8$.

8. Find y

$y = x(x^2 - 4) = (-8)(64 - 4) = -480$, so $P = (-8, -480)$.

Final answer

(a) downward parabola through $(0, 0)$ and $(4, 0)$ with maximum $(2, 20)$.

(b) $f(x) = -5x(x - 4)$.

(c) $P = (-8, -480)$.

Question 10

Integration

10. In this question you must show all stages of your working.

The curve C has equation $y = f(x)$, $x > 0$

Given that

- the point $P(2, 8\sqrt{2})$ lies on C
- $f'(x) = 4\sqrt{x^3} + \frac{k}{x^2}$ where k is a constant
- $f''(x) = 0$ at P

(a) find the exact value of k ,

(4)

(b) find $f(x)$, giving your answer in simplest form.

(4)

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<https://sites.google.com/view/ap-edexcel>

(Total for Question 10 is 8 marks)

Worked Solution - Question 10

1. Differentiate f prime

$$f'(x) = 4x^{3/2} + kx^{-2}, \text{ so } f''(x) = 6x^{1/2} - 2kx^{-3}.$$

2. Use f double prime equals zero at P

$$\text{At } P, x = 2 \text{ and } f''(2) = 0, \text{ so } 6\sqrt{2} - 2k(2^{-3}) = 0.$$

3. Find k

$$6\sqrt{2} - \frac{k}{4} = 0, \text{ hence } k = 24\sqrt{2}.$$

4. Integrate f prime

$$f(x) = \int (4x^{3/2} + 24\sqrt{2}x^{-2}) dx = \frac{8}{5}x^{5/2} - 24\sqrt{2}x^{-1} + c.$$

5. Use the point P

$$P(2, 8\sqrt{2}) \text{ lies on } C, \text{ so } 8\sqrt{2} = \frac{8}{5}(2^{5/2}) - \frac{24\sqrt{2}}{2} + c.$$

6. Find c

$$8\sqrt{2} = \frac{32\sqrt{2}}{5} - 12\sqrt{2} + c, \text{ so } c = \frac{68\sqrt{2}}{5}.$$

Final answer

$$(a) k = 24\sqrt{2}.$$

$$(b) f(x) = \frac{8}{5}x^{5/2} - \frac{24\sqrt{2}}{x} + \frac{68\sqrt{2}}{5}.$$